Scenario 1

```
Flow installed: 00:00:00:00:00:01 → port 1 on sw 1
Flow installed: 00:00:00:00:00:02 → port 2 on sw 1
Flow installed: 00:00:00:00:00:02 → port 2 on sw 1
Flow installed: 00:00:00:00:00:02 → port 2 on sw 1
Flow installed: 00:00:00:00:00:02 → port 2 on sw 1
Flow installed: 00:00:00:00:00:02 → port 2 on sw 1

============================================================
  NETWORK UTILIZATION — Switch 1 — 12:55:57
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          37001723       115284      0.083      0.001     0.8%
  2            119327     36997590      0.001      0.083     0.8%
  3              5759         1626      0.000      0.000     0.0%
============================================================


============================================================
  NETWORK UTILIZATION — Switch 1 — 12:56:02
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          43021389       160230      9.631      0.072    96.3%
  2            164273     43017256      0.072      9.631    96.3%
  3              5759         1626      0.000      0.000     0.0%
============================================================

Flow installed: 00:00:00:00:00:01 → port 1 on sw 1
Flow installed: 00:00:00:00:00:03 → port 3 on sw 1

============================================================
  NETWORK UTILIZATION — Switch 1 — 12:56:07
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          49060583       205122      9.659      0.072    96.6%
  2            202199     47863342      0.061      7.751    77.5%
  3             12725      1229750      0.011      1.964    19.6%
============================================================


============================================================
  NETWORK UTILIZATION — Switch 1 — 12:56:12
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          55073039       248064      9.618      0.069    96.2%
  2            221207     49133422      0.030      2.032    20.3%
  3             36659      5972126      0.038      7.586    75.9%
============================================================
```

```
============================================================
  NETWORK UTILIZATION — Switch 1 — 12:56:17
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          61098749       291162      9.636      0.069    96.4%
  2            240413     50416732      0.031      2.052    20.5%
  3             60551     10730520      0.038      7.609    76.1%
============================================================


============================================================
  NETWORK UTILIZATION — Switch 1 — 12:56:22
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          67121431       334194      9.632      0.069    96.3%
  2            259553     51695632      0.031      2.045    20.5%
  3             84443     15470024      0.038      7.580    75.8%
============================================================


============================================================
  NETWORK UTILIZATION — Switch 1 — 12:56:27
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          73154315       377292      9.649      0.069    96.5%
  2            278693     52974532      0.031      2.045    20.5%
  3            108401     20213740      0.038      7.587    75.9%
============================================================


============================================================
  NETWORK UTILIZATION — Switch 1 — 12:56:32
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          79162607       407850      9.609      0.049    96.1%
  2            284831     53372140      0.010      0.636     6.4%
  3            132821     25821462      0.039      8.969    89.7%
============================================================


============================================================
  NETWORK UTILIZATION — Switch 1 — 12:56:37
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          84075945       427782      7.859      0.032    78.6%
  2            284901     53372140      0.000      0.000     0.0%
  3            152753     30701298      0.032      7.805    78.1%
```

```
=== Network Utilization Monitor Topology Started ===
Hosts: h1(10.0.0.1), h2(10.0.0.2), h3(10.0.0.3)
Run 'pingall' to verify connectivity
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
```

scenario 2

```
  NETWORK UTILIZATION - Switch 1 - 13:10:12
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          84076859       428426      0.000      0.000     0.0%
  2            285465     53372280      0.000      0.000     0.0%
  3            153667     30701942      0.000      0.000     0.0%
============================================================


============================================================
  NETWORK UTILIZATION - Switch 1 - 13:10:17
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          84076859       428426      0.000      0.000     0.0%
  2            285465     53372280      0.000      0.000     0.0%
  3            153667     30701942      0.000      0.000     0.0%
============================================================


============================================================
  NETWORK UTILIZATION - Switch 1 - 13:10:22
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          84076859       428426      0.000      0.000     0.0%
  2            285465     53372280      0.000      0.000     0.0%
  3            153667     30701942      0.000      0.000     0.0%
============================================================


============================================================
  NETWORK UTILIZATION - Switch 1 - 13:10:27
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          84076859       428426      0.000      0.000     0.0%
  2            285465     53372280      0.000      0.000     0.0%
  3            153667     30701942      0.000      0.000     0.0%
============================================================


============================================================
  NETWORK UTILIZATION - Switch 1 - 13:10:32
============================================================
  Port       TX Bytes     RX Bytes    TX Mbps    RX Mbps    Util%
  ------------------------------------------------------------
  1          84076859       428426      0.000      0.000     0.0%
  2            285465     53372280      0.000      0.000     0.0%
  3            153667     30701942      0.000      0.000     0.0%
============================================================
```

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> X h3
h2 -> X X
h3 -> h1 X
*** Results: 66% dropped (2/6 received)
mininet> link s1 h2 up
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3
h2 -> h1 h3
h3 -> h1 h2
*** Results: 0% dropped (6/6 received)
mininet>
```

scenario3

```
vishal@vishal:~/network-utilization-monitor$ sudo ovs-ofctl dump-flows s1
[sudo] password for vishal:
 cookie=0x0, duration=1501.563s, table=0, n_packets=77, n_bytes=5826, priority=0 actions=CONTROLLER:6
5535
vishal@vishal:~/network-utilization-monitor$ sudo ovs-ofctl dump-ports s1
OFPST_PORT reply (xid=0x2): 4 ports
  port LOCAL: rx pkts=0, bytes=0, drop=55, errs=0, frame=0, over=0, crc=0
           tx pkts=0, bytes=0, drop=0, errs=0, coll=0
  port  "s1-eth1": rx pkts=6495, bytes=428986, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=6852, bytes=84078327, drop=0, errs=0, coll=0
  port  "s1-eth2": rx pkts=4379, bytes=53373706, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=4332, bytes=289183, drop=0, errs=0, coll=0
  port  "s1-eth3": rx pkts=2461, bytes=30702502, drop=0, errs=0, frame=0, over=0, crc=0
           tx pkts=2320, bytes=155135, drop=0, errs=0, coll=0
vishal@vishal:~/network-utilization-monitor$
```

```
mininet> h1 iperf -s &
mininet> h2 iperf -c 10.0.0.1 -t 10
------------------------------------------------------------
Client connecting to 10.0.0.1, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.2 port 59612 connected with 10.0.0.1 port 5001
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.4244 sec  11.9 MBytes  9.56 Mbits/sec
mininet>
```